UNIT 2 — 20 QUESTIONS

Feature Detection & Segmentation

Q21. Which of the following is NOT a standard step of the Canny edge detector?

A. Gaussian smoothing
B. Gradient computation
C. Non-maximum suppression
D. Eigenspace projection

Q22. Non-maximum suppression in edge detection is used to:

A. Thicken detected edges
B. Thin edges to single-pixel width
C. Remove colour information
D. Increase image contrast

Q23. The Harris corner detector identifies corners based on:

A. Histogram of pixel values
B. Eigenvalues of the local structure (autocorrelation) matrix
C. Fourier magnitude spectrum
D. Watershed flooding

Q24. In the Harris detector, a true corner region is indicated by:

A. Both eigenvalues small
B. One eigenvalue large, one small
C. Both eigenvalues large
D. Both eigenvalues negative

Q25. SIFT stands for:

A. Scale Invariant Feature Transform
B. Statistical Image Feature Tracker
C. Segmented Interest Feature Type
D. Spatial Intensity Filter Technique

Q26. SIFT descriptors are specifically designed to be invariant to:

A. Colour only
B. Scale and rotation
C. Compression format
D. Frame rate

Q27. HOG stands for:

A. Histogram of Gradients
B. Histogram of Oriented Gradients
C. High Order Gradients
D. Homogeneous Object Grouping

Q28. HOG features are classically well known for their use in:

A. Colour correction
B. Pedestrian/human detection
C. Camera calibration
D. Video compression

Q29. Morphological erosion applied to a binary object generally:

A. Grows the object

B. Shrinks the object
C. Rotates the object
D. Has no effect

Q30. Morphological dilation applied to a binary object generally:
A. Shrinks the object
B. Grows the object
C. Blurs colour channels
D. Removes noise only

Q31. The morphological "opening" operation is defined as:
A. Dilation followed by erosion
B. Erosion followed by dilation
C. Two successive dilations
D. Two successive erosions

Q32. The morphological "closing" operation is defined as:
A. Erosion followed by dilation
B. Dilation followed by erosion
C. Two successive erosions
D. Two successive dilations

Q33. The watershed segmentation algorithm treats a grayscale image as:
A. A frequency spectrum
B. A topographic surface to be flooded
C. A binary mask
D. A feature descriptor vector

Q34. Marker-based watershed segmentation is primarily used to reduce:
A. Computation time
B. Over-segmentation
C. Colour distortion
D. Lens distortion

Q35. Region growing segmentation begins from:
A. Randomly selected pixels only
B. Seed points
C. The image border
D. The frequency spectrum

Q36. Region splitting and merging segmentation is commonly implemented using a:
A. Kalman filter
B. Quad-tree structure
C. Fourier transform
D. Eigenspace

Q37. In graph-based image segmentation, pixels (or regions) are typically represented as:
A. Eigenvectors
B. Nodes of a graph
C. Rows of a histogram
D. Kernel weights

Q38. The Normalized Cut segmentation criterion is designed to:
A. Maximize the total cut weight

B. Balance segment size while minimizing the cut
C. Detect corners only
D. Estimate camera intrinsics

Q39. Mean-shift segmentation clusters pixels by finding:
A. Edges of the image
B. Modes (peaks) of the local density function
C. Vanishing points
D. Kalman gain values

Q40. An active contour ("snake") evolves over iterations to:
A. Maximize an energy function
B. Minimize an energy function
C. Maximize image entropy
D. Randomly reshape itself